

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

3rd Semester of B. Pharm (Old) Examination

December 2015

PH203 Organic Chemistry-I

Date: 07.12.2015, Monday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 80

Instructions:

1. Figures to right indicate marks.
2. Section wise separate answer book to be used.
3. Draw neat sketches and structures, wherever necessary.

SECTION- I

Q1	Attempt any <u>TWO</u> of the following;	20
A	I. Write a note on Kjeldahl's method with suitable diagramme.	06
	II. What is Hybridiazation? Explain SP ² Hybridization in brief.	04
B	I. Explain Inductive and Resonance Effect by taking suitable example.	06
	II. Define Following terms; (Any Two) (02 X 02 = 04 Marks)	04
	a. Intra and Inter molecular Bonding	
	b. Dipole Moment	
	c. Tautomerism	
C	I. Write a brief note on Dumas methods with diagramme.	06
	II. Write down any two preparations and reactions of alkyl halides.	04
Q2	Attempt any <u>TWO</u> of the following;	20
A	Write down reaction, mechanism, kinetic and stereochemistry of SN ₁ and SN ₂ reaction.	10
B	Explain Factors affecting SN ₁ and SN ₂ reactions.	10
C	I. Explain preparations and reactions of Alcohol.	06
	II. Write a short note on Cycloalkane.	04

SECTION- II

Q 3	Attempt any <u>TWO</u> of the following;	20
A	Discuss in details about reaction, mechanism, kinetic, reactivity and orientation in E ₁ and E ₂ mechanism.	10
B	I. Explain any three preparations of alkane and alkene.	05
	II. Explain Saytzeff's and Markonikov's rule with suitable example.	05
C	I. Write preparations and reactions of Ether.	05
	II. Write preparations of Epoxide.	02
	III. Classify dienes with structures.	03
Q 4	Attempt any <u>TWO</u> of the following;	20
A	I. What is Conformation? Explain the conformation in n- Butane with energy diagram.	06
	II. Discuss the sequence rules for specification of absolute configuration.	04
B	Define following terms with examples (Any Five) (05 X 02 = 10)	10
	a. Cis and Trans	
	b. Racemic Mixture	
	c. Meso Compounds	
	d. Optical Activity	
	e. Relative Configuration (D & L)	
	f. E and Z configuration	
	g. Enantiomers	
C	I. Explain stereochemistry in biphenyl system.	05
	II. Enumerate various methods for resolution of racemic mixtures.	05
